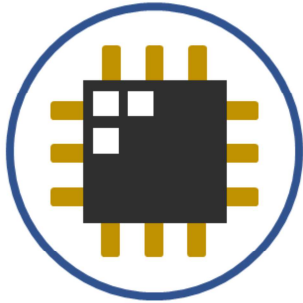




EEE 143: Switching Theory and Digital Logic Design

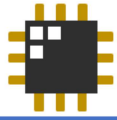
Lecture 7: Combinational circuit non-idealities

1

Last week, we covered encoders, decoders, multiplexers, and demultiplexers, gradually moving up the abstraction level. Instead of just looking at logic gates and truth tables, we're now focusing on how bigger parts of a circuit work together.

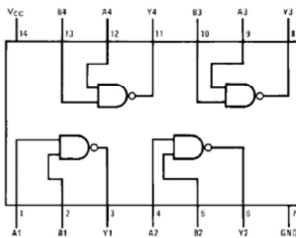
As circuits get more complex, using only truth tables and Karnaugh maps can take too much time. That's why we're now looking at things like adders, subtractors, and comparators—focusing more on what they do rather than just their truth tables.

Now that we're done with combinational logic, the next step is learning about sequential systems. But before we move on, remember that the logic gates we've studied are ideal in theory, but in real life, they have non-idealities.

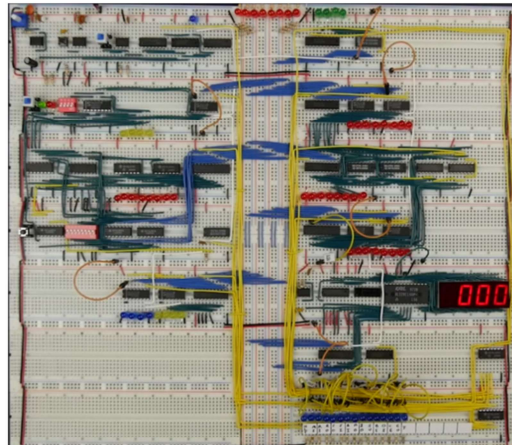


Circuits in real life

Logic gates are actual electrical components!



7400 (Quad 2-input NAND)

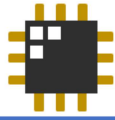


8-bit CPU control logic: Part 1 (Ben Eater)
<https://www.youtube.com/watch?v=dXdoim96v5A>

2

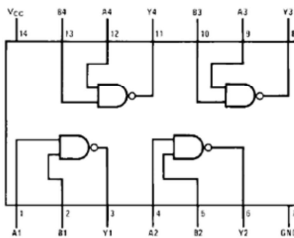
On the left side, you see the 7400 Integrated Circuit (IC), which is a quad 2-input NAND gate, meaning it has four independent NAND gates, each with two inputs. It's part of the 74xx series of TTL (transistor-transistor logic) ICs. The IC has 14 pins, where four pairs of inputs and outputs correspond to the four NAND gates, and two pins for power, which are the Vcc and GND. It is used in basic logic circuits, combinational logic, and even as a universal gate since NAND gates can build any logic function.

On the right, you can see a processor implemented on a breadboard. Since microprocessors are digital systems, they are built by connecting various logic gates, logic blocks, flip-flops, and other components to work together.



Circuits in real life

Real circuits are not ideal... **Datasheet for 7400 (Quad 2-input NAND)**



Electrical Characteristics

over recommended operating free air temperature range (unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ (Note 2)	Max	Units
V_I	Input Clamp Voltage	$V_{CC} = \text{Min}, I_I = -12 \text{ mA}$			-1.5	V
V_{OH}	HIGH Level Output Voltage	$V_{CC} = \text{Min}, I_{OH} = \text{Max}$ $V_{IL} = \text{Max}$	2.4	3.4		V
V_{OL}	LOW Level Output Voltage	$V_{CC} = \text{Min}, I_{OL} = \text{Max}$ $V_{IH} = \text{Min}$		0.2	0.4	V
I_I	Input Current @ Maximum Input Voltage	$V_{CC} = \text{Max}, V_I = 5.5 \text{ V}$			1	mA
I_{IH}	HIGH Level Input Current	$V_{CC} = \text{Max}, V_I = 2.4 \text{ V}$			40	μA
I_{IL}	LOW Level Input Current	$V_{CC} = \text{Max}, V_I = 0.4 \text{ V}$			-1.6	mA
I_{OS}	Short Circuit Output Current	$V_{CC} = \text{Max}$ (Note 3)	-18		-55	mA
I_{OCH}	Supply Current with Outputs HIGH	$V_{CC} = \text{Max}$		4	8	mA
I_{OCL}	Supply Current with Outputs LOW	$V_{CC} = \text{Max}$		12	22	mA

Note 2: All typicals are at $V_{CC} = 5 \text{ V}$, $T_A = 25^\circ\text{C}$.

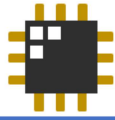
Note 3: Not more than one output should be shorted at a time.

Switching Characteristics

at $V_{CC} = 5 \text{ V}$ and $T_A = 25^\circ\text{C}$

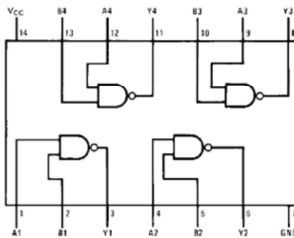
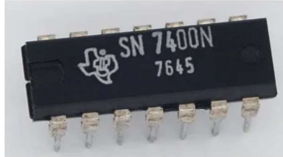
Symbol	Parameter	Conditions	Min	Max	Units
t_{PLH}	Propagation Delay Time LOW-to-HIGH Level Output	$C_L = 15 \text{ pF}$ $R_L = 400\Omega$		22	ns
t_{PHL}	Propagation Delay Time HIGH-to-LOW Level Output			15	ns

Since these are electrical components, they have specific electrical characteristics, such as the required **voltage supply** needed for the IC to function and the **current it draws** from the power source. These characteristics determine how the IC operates within a circuit and ensure it works reliably without damage.



Circuits in real life

Real circuits are not ideal...



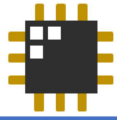
For this lecture, we will focus on delays and how they affect the circuit specifications of our design

Switching Characteristics

at $V_{CC} = 5V$ and $T = 25^\circ C$

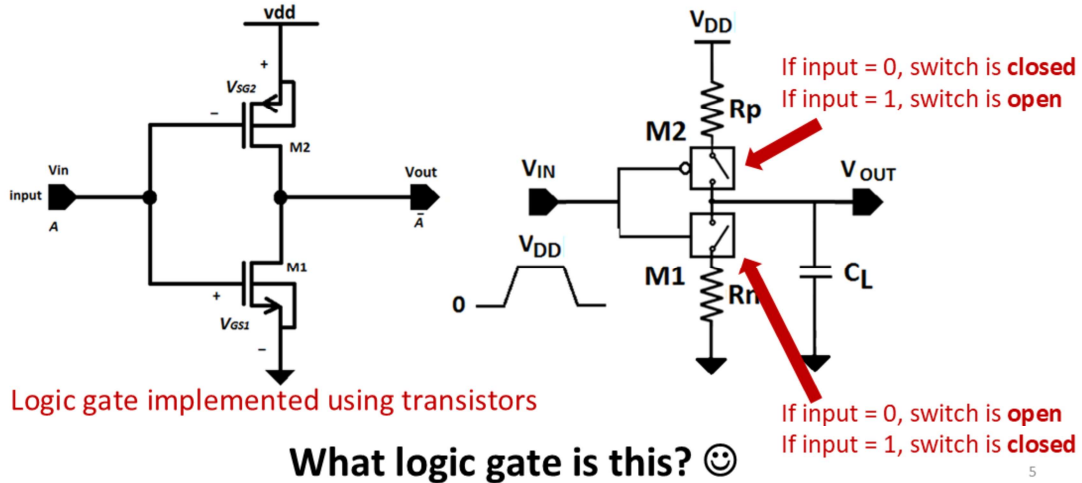
Symbol	Parameter	Conditions	Min	Max	Units
t_{PLH}	Propagation Delay Time LOW-to-HIGH Level Output	$C_L = 15\text{ pF}$ $R_L = 400\Omega$		22	ns
t_{PHL}	Propagation Delay Time HIGH-to-LOW Level Output			15	ns

Now, let's move on to **switching characteristics**, which focus on **delay**. When you provide an input to a logic gate, the output **does not appear instantly**, it takes a small amount of time, usually in the **nanosecond range**. In this lecture, we will focus on how **delay in real-life logic gates** impacts system performance. However, we won't go into much detail about **electrical characteristics**.



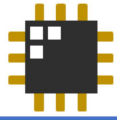
Circuits in real life

Logic gates are implemented using transistors in real life



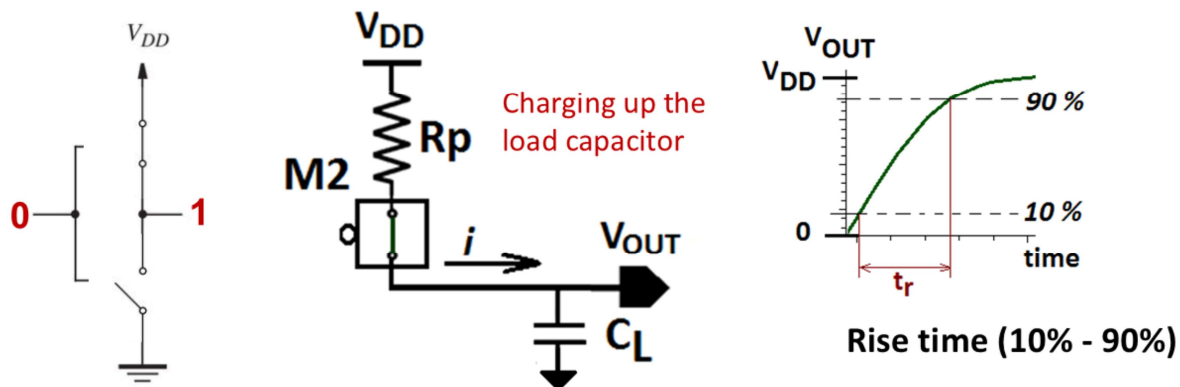
Here, we are showing how a **CMOS inverter** is implemented using **transistors**, which act as **switches**. In CMOS technology, there are two types of transistors: **PMOS (normally OFF)** and **NMOS (normally ON)**. When connected together, they create the **switch-level equivalent** shown on the right.

When the **input is 1**, the **NMOS transistor turns ON** (acting as a closed switch), while the **PMOS transistor turns OFF** (acting as an open switch). However, these transistors are not just ideal switches—they also have **resistance** and **parasitic capacitance**. This means that every time a transistor switches states, it introduces a **resistance (R)**, and due to capacitance (C), the circuit behaves like an **RC network**. This RC effect causes a **delay** in signal transitions, leading to gradual voltage changes rather than instant switching.



Circuits in real life

0-to-1 output delay

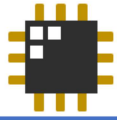


6

Now, to tackle more about the CMOS logic gate, we first look at the 0-to-1 output delay. Since we are focusing on the output, we consider the case where the output transitions to 1.

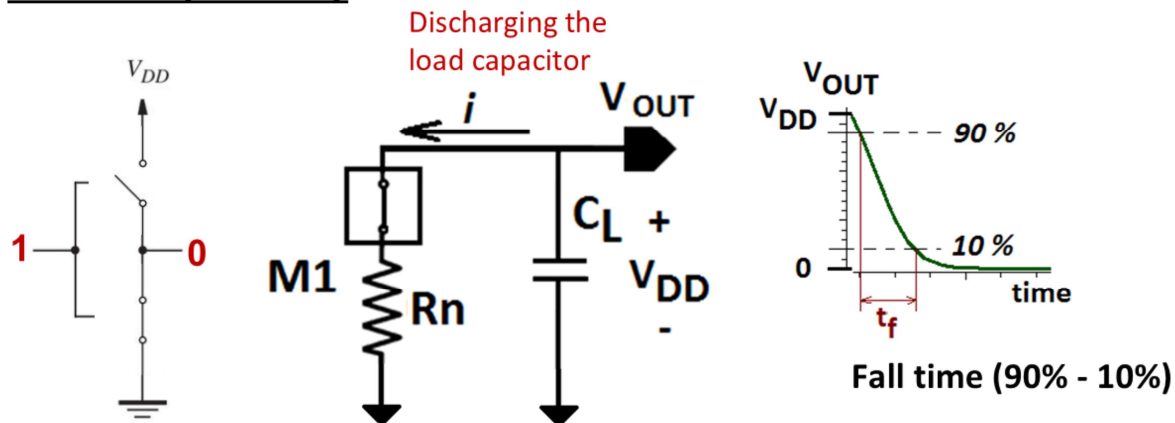
Initially, the **output is 0**. When the **input switches to 0**, it causes the **output to become 1**. Looking at the circuit (left figure), we see that the **PMOS transistor is conducting (shorted)** while the **NMOS transistor is off (open)**. This means the **load capacitance is now connected to Vdd**, forming an **RC circuit** with the pull-up resistor (middle figure).

On the right, we see the corresponding **waveform**, which follows an **RC time constant**, illustrating how the output voltage gradually rises instead of switching instantly.



Circuits in real life

1-to-0 output delay

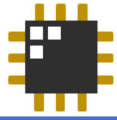


7

Of course, we also have the **1-to-0 output delay**, where we focus on the **output transitioning from 1 to 0**.

Looking at the **left figure**, we see that the **NMOS transistor is conducting (shorted)** while the **PMOS transistor is off (open)**. This means the **load capacitance is now connected to ground** through a resistor, once again forming an **RC circuit** (middle figure).

In the **right figure**, the waveform follows an **RC time constant**, showing how the output **gradually decreases** instead of dropping instantly.



Propagation delay

- The time interval from the triggering edge of the signal to the stabilization of the new state (or output)
- Measured from 50% (input transition) to 50% (output transition)
- 2 types: Low-to-high (t_{pLH}) and high-to-low (t_{pHL}) delays

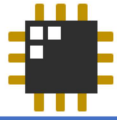
8

Now, we will move on to propagation delay. So, propagation delay simply refers to how fast the output changes after the input changes. You always measure it from the moment the input switches because if the input stays the same, the output remains steady. But once the input changes, the output takes some time to respond, and we want to measure how long that takes.

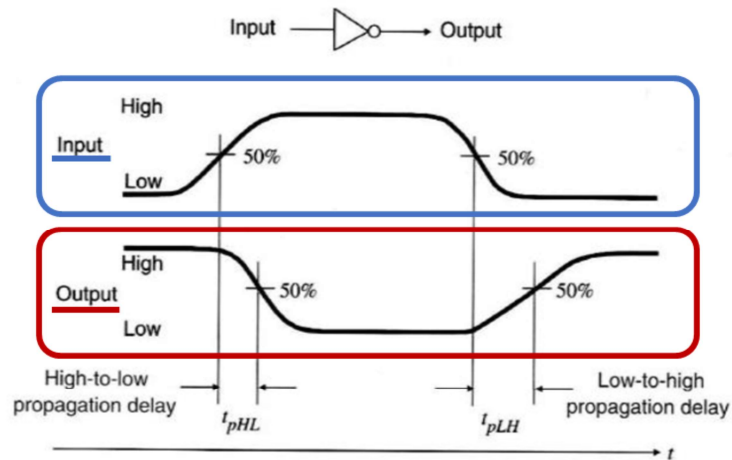
To do this, we typically measure the delay from the point where the input reaches **50% of its transition** to when the output reaches **50% of its transition**. There are **two types of delays**:

- **Low-to-High (t_{pLH}):** When the output goes from **0 to 1**.
- **High-to-Low (t_{pHL}):** When the output goes from **1 to 0**.

These delays help us understand how quickly a circuit reacts to changes in input signals

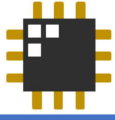


Propagation delay



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Here, we can see the propagation delay in a digital circuit using an inverter. When the input signal (blue box) transitions from **low to high** or **high to low**, the output (red box) responds with the opposite transition, but not instantly. Instead, there is a slight delay before the output stabilizes. This delay is measured at the **50% transition points** of both the input and output signals. The **high-to-low propagation delay (t_{pHL})** represents the time it takes for the output to switch from **high (1) to low (0)** after the input changes, while the **low-to-high propagation delay (t_{pLH})** measures the time taken for the output to switch from **low (0) to high (1)**. These delays occur due to real-world circuit properties like **resistance and capacitance**, which affect how quickly signals can change. Understanding propagation delay is crucial in **high-speed digital system design** to ensure proper timing and synchronization of signals.

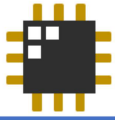


Causes of propagation delay

- Propagation time can also change depending on:
 - **Fan-in:** the number of inputs that a logic gate has
 - More inputs -> more transistors -> more complex circuit -> more delay

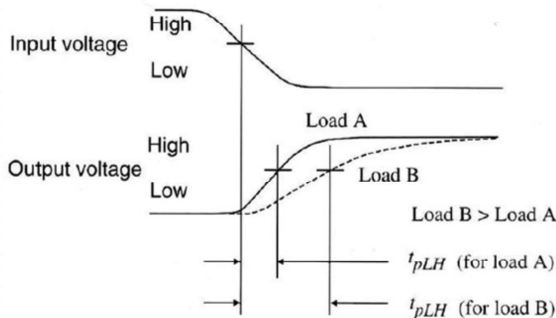
10

Now, we will move on to some of the causes of propagation delay. First, the **fan-in**, which refers to the number of inputs a logic gate has, and as this number increases, the gate's internal structure becomes complex. This happens because each additional input requires more transistors to process the logic operation, which in turn increases the **circuit complexity**. With more transistors involved, the delay in switching states also increases due to additional **capacitance and resistance** within the circuit. This added delay means that a logic gate with many inputs will take longer to produce an output compared to one with fewer inputs.



Causes of propagation delay

- Propagation time can also change depending on:
 - **Fan-in:** the number of inputs that a logic gate has
 - More inputs -> more transistors -> more complex circuit -> more delay
 - **Fan-out:** the number of connections to the output of a logic gate

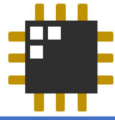


Effect of fan-out:

A logic gate that has more output connections (AKA load) will be slower!

11

Now, we will move to **fan-out**, which is the number of outputs of logic gate needs to drive that also affects propagation delay. **Fan-out refers to how many other gates or components are connected to a gate's output.** When a logic gate has to send its signal to multiple other gates, it experiences a heavier **electrical load**. This happens because each additional connection introduces more **capacitance**, which slows down the rate at which the output voltage can change. As a result, the gate takes longer to switch between logic states, increasing the overall **delay** in the circuit.



Propagation delay

- Example logic gate delay characteristic

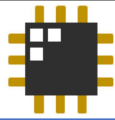
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

L typically refers to the number of connections to the logic gate output*

**Note: Different logic gate inputs can have different loading factors, but we'll assume each input has a load factor of 1 for simplicity*

12

Here, we can see different variations of **AND and OR gates**, each with different **fan-in values** and corresponding **propagation delays**. The equations shown include a variable **L**, which typically represents the number of connections to the output of a logic gate. Additionally, we can observe both **low-to-high (tpLH) and high-to-low (tpHL) propagation delays**, as these two values are not always identical. Another key observation is that **as the number of inputs increases, the delay also increases**, highlighting the impact of circuit complexity on signal timing.



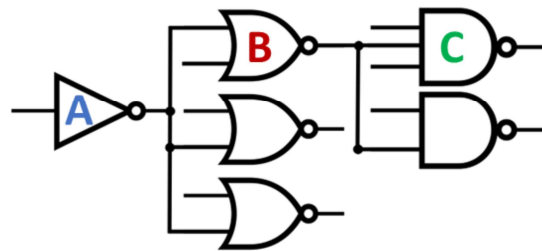
Propagation delay

- Example logic gate delay characteristic

Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

L typically refers to the number of connections to the logic gate output*

**Note: Different logic gate inputs can have different loading factors, but we'll assume each input has a load factor of 1 for simplicity*



A is a NOT gate with **L = 3**

B is a 2-input NOR gate with **L = 2**

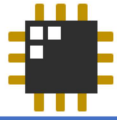
C is a 3-input NAND gate with **L = 1**
(default load is 1 if not explicitly drawn)

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Let's go through this example by tracing the path from **A to B to C** and calculating the total delay from the input to the final output at **C**. As shown in the diagram, the signal passes through **three logic gates**, each contributing to the overall delay.

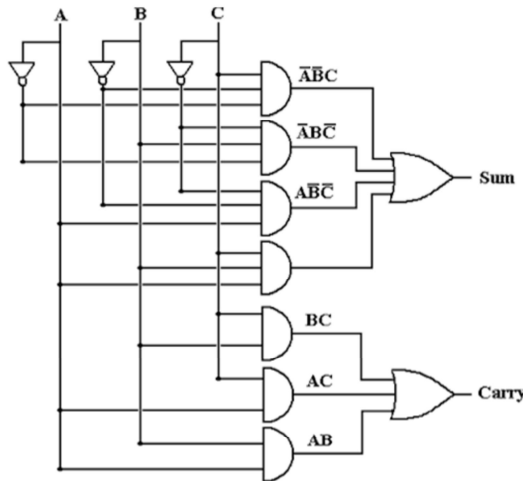
- **At A**, the signal first goes through a **NOT gate** with **L = 3**, meaning its output is connected to three other inputs.
- **At B**, the signal passes through a **2-input NOR gate** with **L = 2**, since its output has two connections.
- **At C**, the final stage is a **3-input NAND gate** with **L = 1**, as it only drives one output.

By considering the propagation delays of each gate, we can determine how long it takes for the signal to travel from **A to C**, factoring in the effects of **fan-in** and **fan-out** at each stage.



Critical path calculation

- Given the following circuit, determine the longest path delay



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

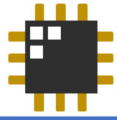
14

Critical path calculation is a way to determine the longest time it takes for a signal to travel from an input to an output in a digital circuit. In simpler terms, it helps us find the slowest path in a circuit, which limits how fast the entire system can operate.

Let's go through this example and identify the path that experiences the most **fan-in** and **fan-out** along the way. Looking at the circuit, we can see that:

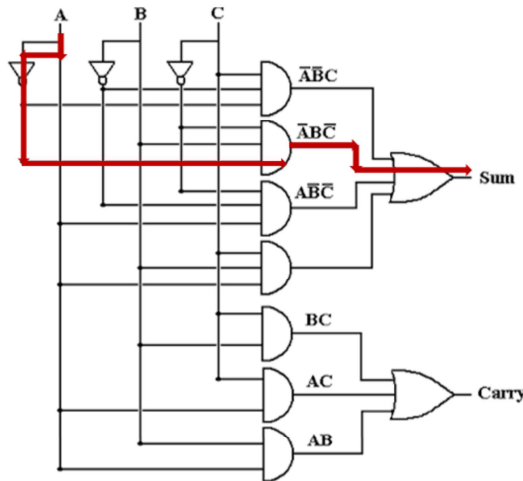
- Each **NOT gate** has **one input** and a fan-out of **L = 2**
- The four **AND gates** at the top have **three inputs** each and **L = 1**
- The **OR gate** at the top has **four inputs** and **L = 1**

By tracing the paths that include this combination of gates, we can determine that they result in the **longest propagation delay**. This is because larger **fan-in** increases gate complexity, and higher **fan-out** adds more load, both of which contribute to longer delays.



Critical path calculation

- Given the following circuit, determine the longest path delay



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

Notation: Logic gate, Fan-in (#inputs)

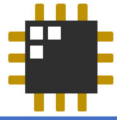
A → INV → AND3 → OR4 → Sum
(this is just one of the longest paths in the circuit)

15

As mentioned in the previous slide, the **longest path delay** occurs when tracing the path that includes the **NOT gate, 3-input AND gate, and 4-input OR gate**. This happens because **higher fan-in** makes the gate more complex, increasing the processing time, while **higher fan-out** adds more load, further slowing down the response. In this case, we selected the path

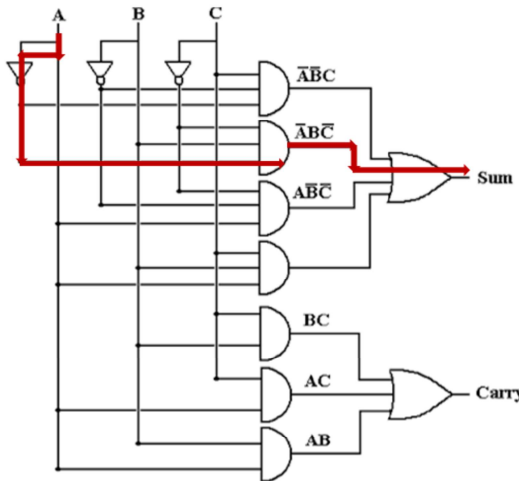
A → NOT gate → 3-input AND gate → 4-input OR gate → SUM as an example,

but it's important to note that two other paths have the same delay. This particular path was chosen for clarity in illustrating the concept.



Critical path calculation

- Given the following circuit, determine the longest path delay



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

If A changes from 1 to 0:

$$\begin{aligned}
 t_{pLH} &= \text{NOT}(LH) + \text{AND3}(LH) + \text{OR4}(LH) \\
 &= 0.02 + 0.038(2) + 0.20 + 0.038(1) + 0.13 + 0.038(1) \\
 t_{pLH} &= \underline{0.502 \text{ ns}}
 \end{aligned}$$

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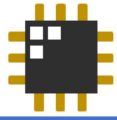
Now that we have identified the longest path in the circuit, we can determine the total propagation delay by referring to the table. Based on our chosen path, the NOT gate has **L = 2**, the AND gate has **3 inputs**, and the OR gate has **4 inputs**.

Since the input transition we are considering is **from 1 to 0**, we need to account for how each gate responds. The first gate in the path is a **NOT gate**, which inverts the signal, meaning its **output transitions from 0 to 1**. Because of this, we must use the **low-to-high (LH) propagation delay** for the NOT gate, which we will also use for the rest of the gates.

Thus, the total propagation delay is calculated as follows:

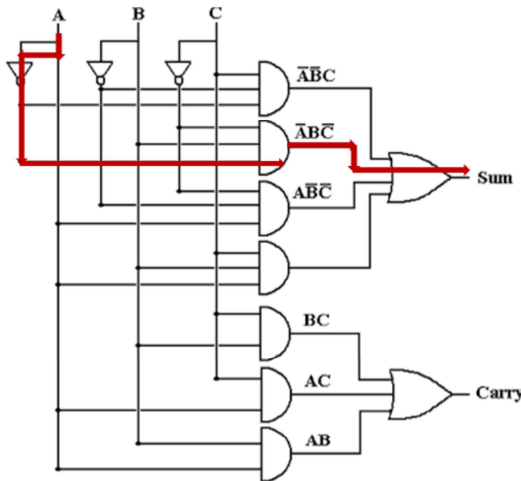
$$\begin{aligned}
 t_{pLH} &= \text{NOT}(LH) + \text{AND3}(LH) + \text{OR4}(LH) \\
 &= 0.02 + 0.038(2) + 0.20 + 0.038(1) + 0.13 + 0.038(1)
 \end{aligned}$$

By summing these values, we obtain a total propagation delay of **0.502 ns** for this longest path.



Critical path calculation

- Given the following circuit, determine the longest path delay



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

If A changes from 0 to 1:

$$\begin{aligned}
 t_{pHL} &= \text{NOT(HL)} + \text{AND3(HL)} + \text{OR4(HL)} \\
 &= 0.05 + 0.017(2) + 0.18 + 0.018(1) + 0.45 + 0.025(1) \\
 t_{pHL} &= \underline{0.757 \text{ ns}}
 \end{aligned}$$

17

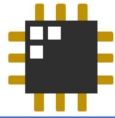
Now, let's analyze the case when **A changes from 0 to 1**. Since the first gate in the path is a **NOT gate**, its output will transition **from 1 to 0**, meaning we need to use the **high-to-low (HL) propagation delay** for this gate.

As before, we consider the **fan-in and fan-out** of each gate along the longest path and refer to the table to determine the corresponding delays. Based on our chosen path, the NOT gate has **L = 2**, the AND gate has **3 inputs**, and the OR gate has **4 inputs**.

Using the provided values, the total **high-to-low propagation delay** is calculated as follows:

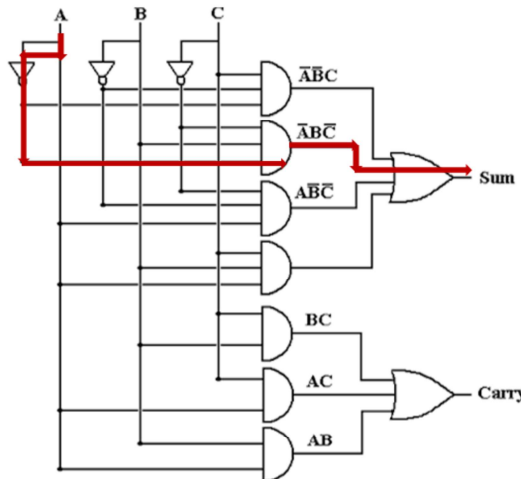
$$\begin{aligned}
 t_{pHL} &= \text{NOT(HL)} + \text{AND3(HL)} + \text{OR4(HL)} \\
 &= 0.05 + 0.017(2) + 0.18 + 0.018(1) + 0.45 + 0.025(1)
 \end{aligned}$$

Thus, the total delay for this transition is 0.757 ns.



Critical path calculation

- Given the following circuit, determine the longest path delay



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

If A changes from 1 to 0:

$$t_{pLH} = 0.502 \text{ ns}$$

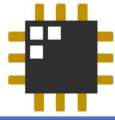
If A changes from 0 to 1:

$$t_{pHL} = 0.757 \text{ ns}$$

Always consider the worst case!

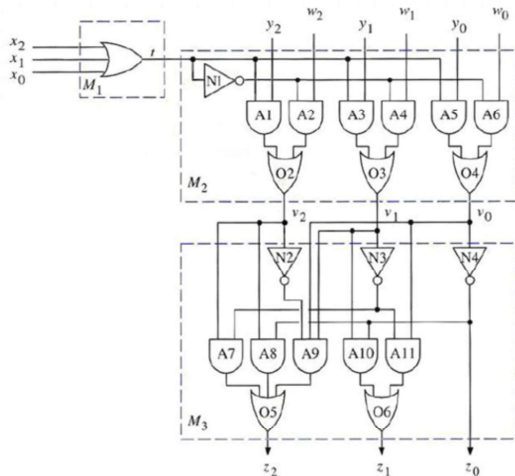
18

Now that we have calculated the **propagation delays** for both **low-to-high (LH)** and **high-to-low (HL)** transitions, we determine the **longest path delay** by selecting the **worst-case scenario**—the delay with the **longest duration**. In this example, the **high-to-low propagation delay** of **0.757 ns** is the largest, making it the **critical delay** for the circuit.



Critical path calculation

- Practice: Determine the longest path delay



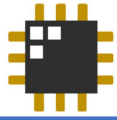
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

19

Again, to determine the longest path delay, we identify the gates with the highest fan-in and fan-out along the way. In this example, the longest path includes the following gates:

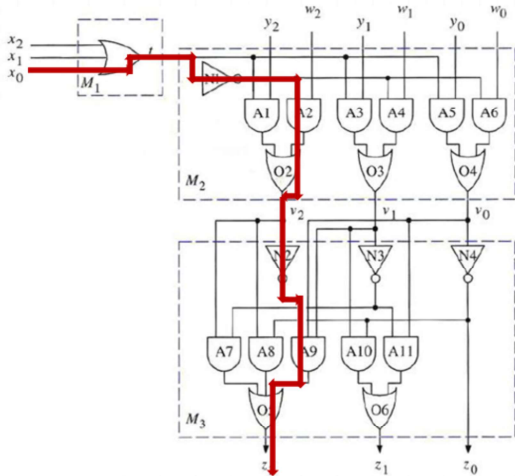
- OR gate with 3 inputs and $L=4$
- NOT gate with 1 input and $L=3$
- AND gate with 2 inputs and $L=1$
- OR gate with 2 inputs and $L=3$
- NOT gate with 1 input and $L=1$
- AND gate with 3 inputs and $L=1$
- OR gate with 3 inputs and $L=1$

These gates collectively determine the longest path delay in the circuit.



Critical path calculation

- Practice: Determine the longest path delay



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

Notation: Logic gate, Fan-in (#inputs)

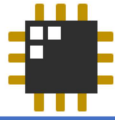
$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$

(this is the only longest path in the circuit)

20

As mentioned in the previous slide, the longest path in this circuit occurs when the input passes through the following sequence of gates: an **OR gate with 3 inputs**, a **NOT gate**, an **AND gate with 2 inputs**, an **OR gate with 2 inputs**, another **NOT gate**, an **AND gate with 3 inputs**, and finally, an **OR gate with 3 inputs** before reaching the output. Therefore, for this example, the longest path delay is determined by the sequence:

$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow \text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$.



Critical path calculation

- Practice: Determine the longest path delay

If A changes from 1 to 0:

$$t_{pHL} = \text{OR3}(\underline{HL}) + \text{NOT}(\underline{LH}) + \text{AND2}(\underline{LH}) + \text{OR2}(\underline{LH}) + \text{NOT}(\underline{HL}) + \text{AND3}(\underline{HL}) + \text{OR3}(\underline{HL})$$

Note how the HL, LH switches whenever we encounter a NOT gate!

Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

Now, we determine the L values per logic gate

$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow \text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$

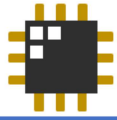
(this is the only longest path in the circuit)

21

Let's start by analyzing the case where **A transitions from 1 to 0**. Since **NOT gates invert the signal**, every time we pass through one, a **low-to-high (LH) transition at the input becomes a high-to-low (HL) transition at the output**, and vice versa. This flipping effect only occurs with **NOT gates**—for other types of gates, the transition type remains the same. Keeping this in mind, we can now determine the total propagation delay for this longest path.

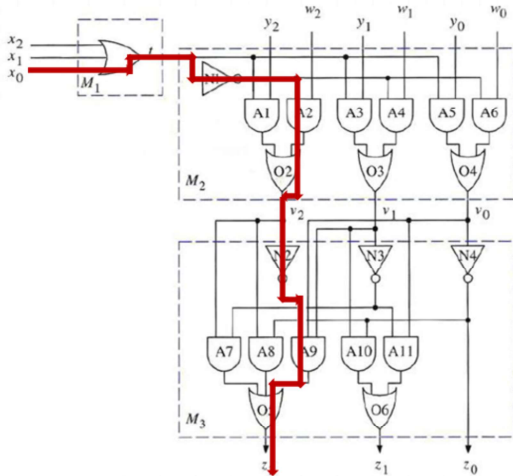
$$t_{pHL} = \text{OR3}(\underline{HL}) + \text{NOT}(\underline{LH}) + \text{AND2}(\underline{LH}) + \text{OR2}(\underline{LH}) + \text{NOT}(\underline{HL}) + \text{AND3}(\underline{HL}) + \text{OR3}(\underline{HL})$$

Now, let's determine the L values per logic gate.



Critical path calculation

- Practice: Determine the longest path delay



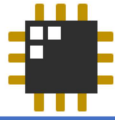
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow \text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$

(this is the only longest path in the circuit)

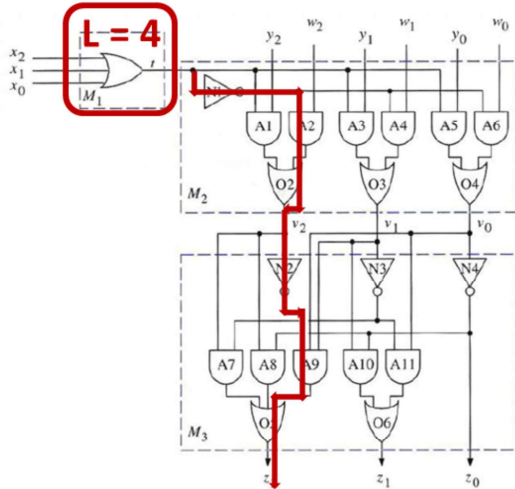
22

We begin with the **3-input OR gate**. By examining the circuit, we can see that its output is connected to **four other gates**, such as NOT gate and three AND gates, meaning the **fan-out (L)** is 4.



Critical path calculation

- Practice: Determine the longest path delay



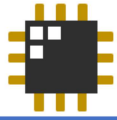
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$

(this is the only longest path in the circuit)

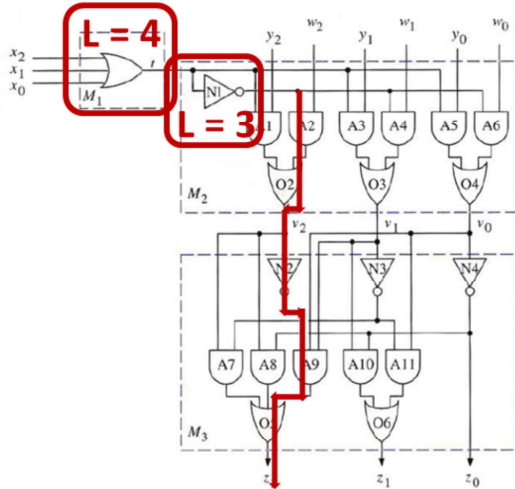
23

Next, we analyze the **NOT gate with 1 input**. Based on the circuit, we can see that its output is connected to **three AND gates**, giving it a **fan-out of $L = 3$** .



Critical path calculation

- Practice: Determine the longest path delay



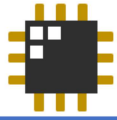
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

**$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$**

(this is the only longest path in the circuit)

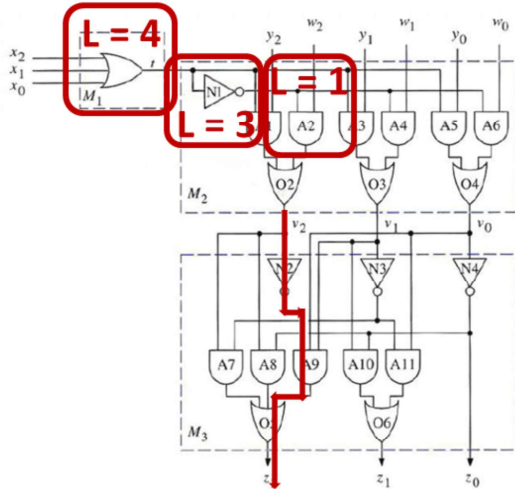
24

Now, moving on to the **2-input AND gate**, we can observe that its output is only connected to a single **OR gate**, meaning its **fan-out is $L = 1$** .



Critical path calculation

- Practice: Determine the longest path delay

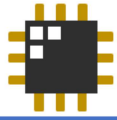


Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

**$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$**
(this is the only longest path in the circuit)

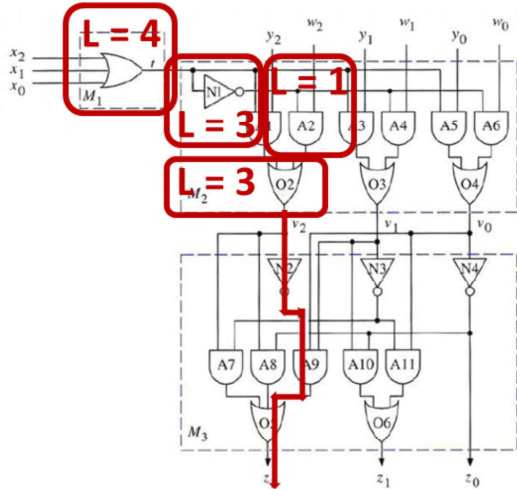
25

Next, let's examine the **2-input OR gate**. Its output is connected to **two AND gates and one NOT gate**, resulting in a **fan-out of L = 3**.



Critical path calculation

- Practice: Determine the longest path delay



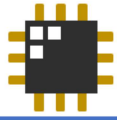
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

**$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$**

(this is the only longest path in the circuit)

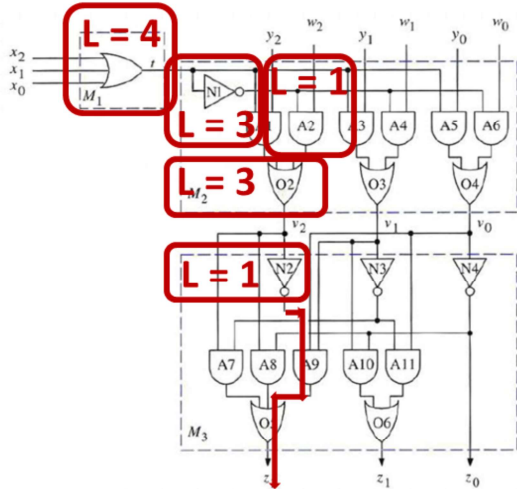
26

Now, looking at the next **NOT gate with 1 input**, its output is connected to only **one AND gate**, so we have **L = 1**.



Critical path calculation

- Practice: Determine the longest path delay



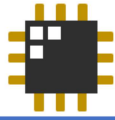
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

**$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$**

(this is the only longest path in the circuit)

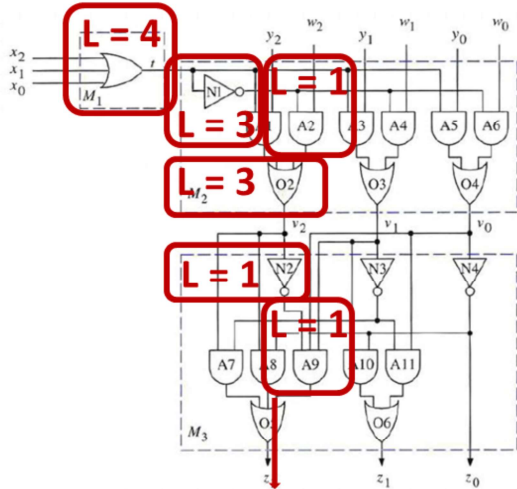
27

Next, for the **AND gate with 3 inputs**, its output is connected only to an **OR gate**, so we have **L = 1**.



Critical path calculation

- Practice: Determine the longest path delay



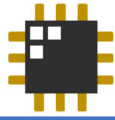
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

**$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$**

(this is the only longest path in the circuit)

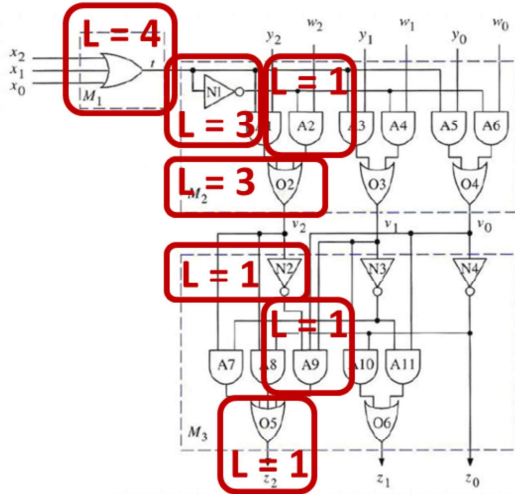
28

Lastly, for the **OR gate with 3 inputs**, since its output is not connected to any other gates, we have **L = 1**.



Critical path calculation

- Practice: Determine the longest path delay



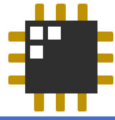
Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

**$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow$
 $\text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$**

(this is the only longest path in the circuit)

29

Now, we found the fan-out (L) for all gates. We will now move on getting the longest path delay using the table on the right.



Critical path calculation

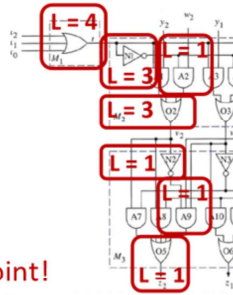
- Practice: Determine the longest path delay

If A changes from 1 to 0:

$$t_{pHL} = \text{OR3}(\underline{HL}) + \text{NOT}(\underline{LH}) + \text{AND2}(\underline{LH}) + \text{OR2}(\underline{LH}) + \text{NOT}(\underline{HL}) + \text{AND3}(\underline{HL}) + \text{OR3}(\underline{HL})$$

$$t_{pHL} = 0.34 + 0.022(4) + 0.02 + 0.038(3) + 0.15 + 0.037(1) + 0.12 + 0.037(3) + 0.05 + 0.017(1) + 0.18 + 0.018(1) + 0.34 + 0.022(1)$$

Just perform the computation at this point!
Do the same for t_{pLH} !



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow \text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$

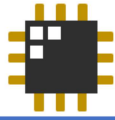
(this is the only longest path in the circuit)

30

Using the fan-in and fan-out values we previously determined, we can now refer to the table on the right to compute the longest path delay. Hence, we will have:

$$t_{pHL} = 0.34 + 0.022(4) + 0.02 + 0.038(3) + 0.15 + 0.037(1) + 0.12 + 0.037(3) + 0.05 + 0.017(1) + 0.18 + 0.018(1) + 0.34 + 0.022(1)$$

We get a value of 1.607 ns for high-to-low propagation delay.



Critical path calculation

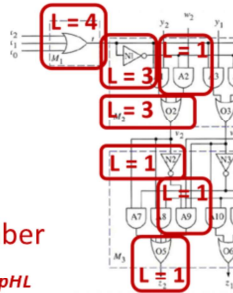
- Practice: Determine the longest path delay

If A changes from 0 to 1:

$$t_{pLH} = \text{OR3}(\underline{\text{LH}}) + \text{NOT}(\underline{\text{HL}}) + \text{AND2}(\underline{\text{HL}}) + \text{OR2}(\underline{\text{HL}}) + \text{NOT}(\underline{\text{LH}}) + \text{AND3}(\underline{\text{LH}}) + \text{OR3}(\underline{\text{LH}})$$

$$t_{pLH} = 0.12 + 0.038(4) + 0.05 + 0.017(3) + 0.16 + 0.017(1) + 0.20 + 0.019(3) + 0.02 + 0.038(1) + 0.20 + 0.038(1) + 0.12 + 0.038(1)$$

Pick the bigger number between t_{pHL} and t_{pLH} to get the worst case delay!



Gate Type	Fan-in	Propagation Delays	
		t_{pLH} (ns)	t_{pHL} (ns)
AND	2	$0.15 + 0.037L$	$0.16 + 0.017L$
AND	3	$0.20 + 0.038L$	$0.18 + 0.018L$
AND	4	$0.28 + 0.039L$	$0.21 + 0.019L$
OR	2	$0.12 + 0.037L$	$0.20 + 0.019L$
OR	3	$0.12 + 0.038L$	$0.34 + 0.022L$
OR	4	$0.13 + 0.038L$	$0.45 + 0.025L$
NOT	1	$0.02 + 0.038L$	$0.05 + 0.017L$

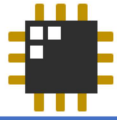
$x_i \rightarrow \text{OR3} \rightarrow \text{INV} \rightarrow \text{AND2} \rightarrow \text{OR2} \rightarrow \text{INV} \rightarrow \text{AND3} \rightarrow \text{OR3} \rightarrow z_i$
(this is the only longest path in the circuit)

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Now, let's analyze the case when A changes from 0 to 1. Using the corresponding propagation delay values from the table, we calculate:

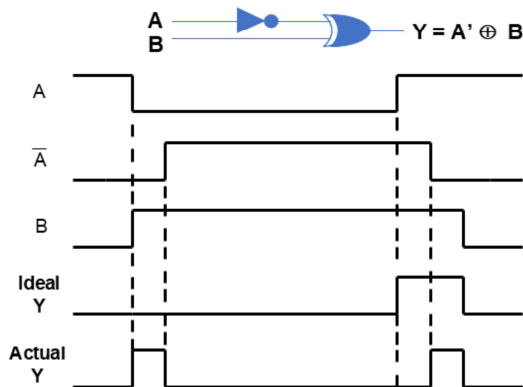
$$t_{pLH} = 0.12 + 0.038(4) + 0.05 + 0.017(3) + 0.16 + 0.017(1) + 0.20 + 0.019(3) + 0.02 + 0.038(1) + 0.20 + 0.038(1) + 0.12 + 0.038(1)$$

We get 1.261 ns for the low-to-high propagation delay. Hence the critical delay is 1.607 ns.



Glitches

- Because of these propagation delays, momentary glitches may occur at the final output as each logic gate stabilizes
 - Make sure you consider the additive delays along the different paths in your circuit

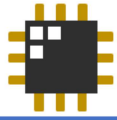


The longest path will determine how fast we can get the correct output!

32

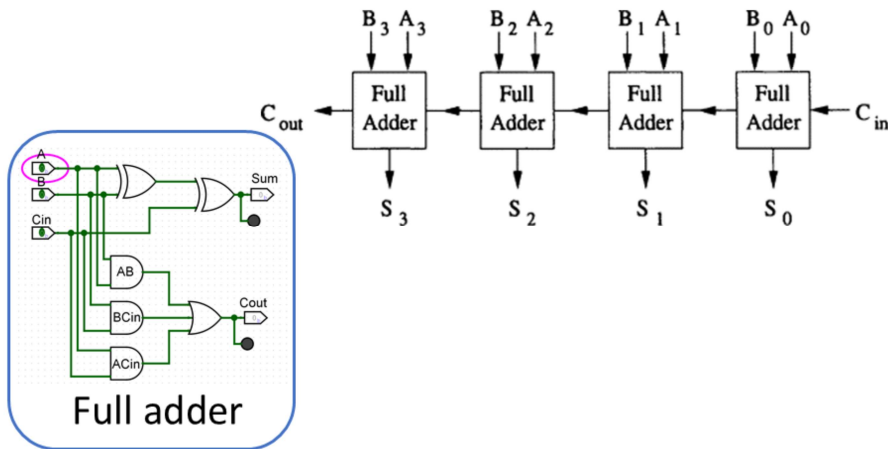
Due to propagation delays, brief glitches can appear at the output before the circuit fully stabilizes. These temporary fluctuations happen because different logic paths take varying amounts of time to process signals. It's crucial to account for these cumulative delays throughout the circuit, as they impact timing and reliability. Ultimately, the longest path dictates the minimum time required to produce a stable and accurate output.

Refer to the example on the left, where A transitions from 1 to 0 while B is 1. As A passes through the NOT gate, its output doesn't update instantly. Because of this delay, when XORed with B, Y temporarily holds a value of 1 before settling to the expected 0. Similarly, when A transitions from 0 to 1 while B remains 1, the NOT gate's delayed response causes Y to still be 0 instead of immediately switching to the correct value of 1.



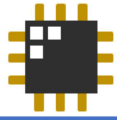
Adder propagation delay

- Given a 4-bit ripple carry adder, what's the **longest path** to get the correct sum $S_3S_2S_1S_0$?



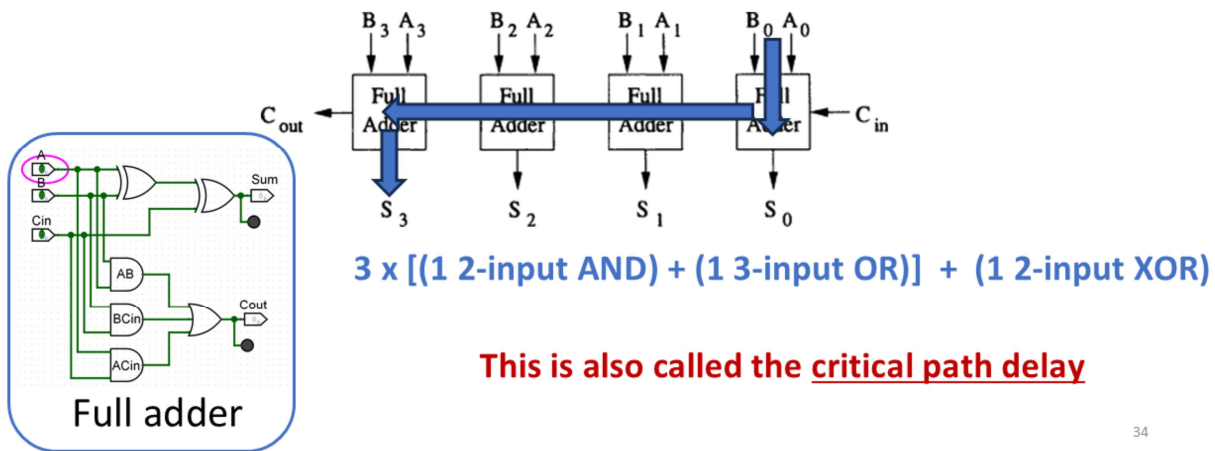
33

In a ripple carry adder, each full adder relies on the carry output of the previous stage. This means that the longest delay occurs when the carry signal propagates through all full adders, from the least significant bit (LSB) to the most significant bit (MSB).



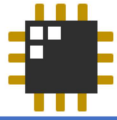
Adder propagation delay

- Given a 4-bit ripple carry adder, what's the **longest path** to get the correct sum $S_3S_2S_1S_0$?



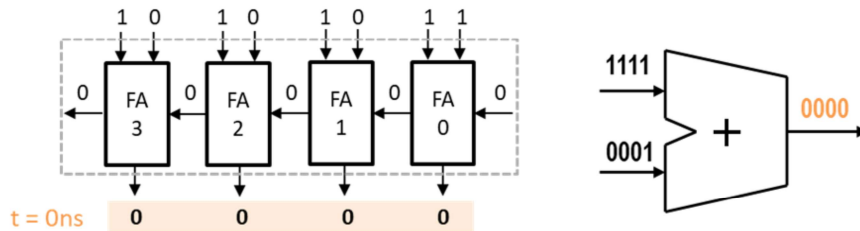
34

The delay in a 4-bit ripple carry adder is primarily determined by the logic gates involved in propagating the carry and computing the sum. Each full adder consists of **AND, OR, and XOR gates**, each contributing to the total propagation delay. The equation given on the slide breaks this down into three stages, where each stage consists of a **2-input AND gate** and a **3-input OR gate** to handle carry propagation. Since the carry must pass through multiple full adders, this delay accumulates over three stages. Finally, the sum computation introduces an additional delay due to a **2-input XOR gate**, which is required to generate the final sum bit. The cumulative delay from these gates defines the **critical path**, meaning that the time taken for the signal to travel through all necessary gates sets the speed limit of the circuit.



Timing considerations

- We often want a circuit to be as fast as possible. However, it takes some time for signals to propagate (propagation delay)

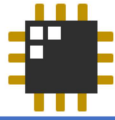


Inputs changed but each FA takes some time to 'react'

* For simplicity, assume sum and carry-out are output by each FA after 1ns from input change

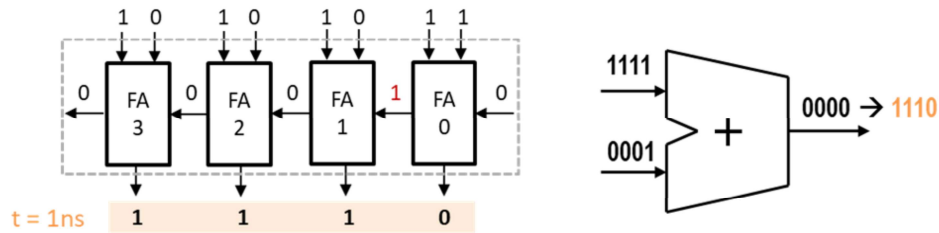
35

In this example, we are using the same circuit as before, but now we focus on understanding what to expect during operation. Imagine applying the inputs **1111 + 0001** simultaneously on the adder side. As we analyze the behavior over time, we can observe how the system processes these inputs at the nanosecond level. The key takeaway is to visualize the **glitches** that occur due to propagation delays. Initially, at $t = 0\text{ns}$, no changes have taken effect yet, so the output remains **0000**.



Timing considerations

- We often want a circuit to be as fast as possible. However, it takes some time for signals to propagate (propagation delay)



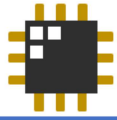
Outputs updated based on inputs but sum is wrong!

Carries have not propagated

* For simplicity, assume sum and carry-out are output by each FA after 1ns from input change

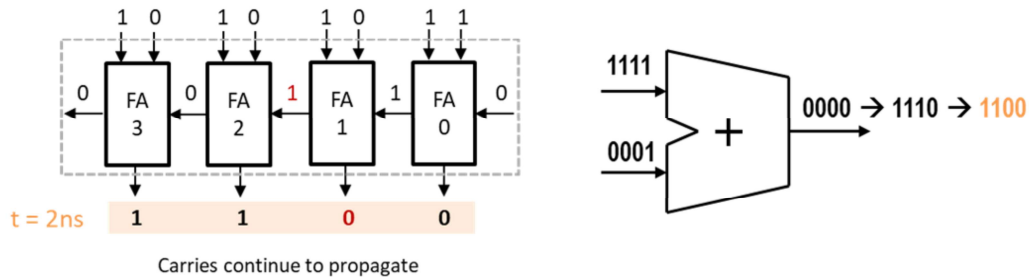
36

At **t = 1ns**, the output of **FA0** is **0**, with **Cin** equal to **1**. However, it's important to note that the addition in **FA1**, **FA2**, and **FA3** occurs simultaneously with **FA0**. As a result, their outputs collectively form **1110** at this moment.



Timing considerations

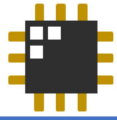
- We often want a circuit to be as fast as possible. However, it takes some time for signals to propagate (propagation delay)



* For simplicity, assume sum and carry-out are output by each FA after 1ns from input change

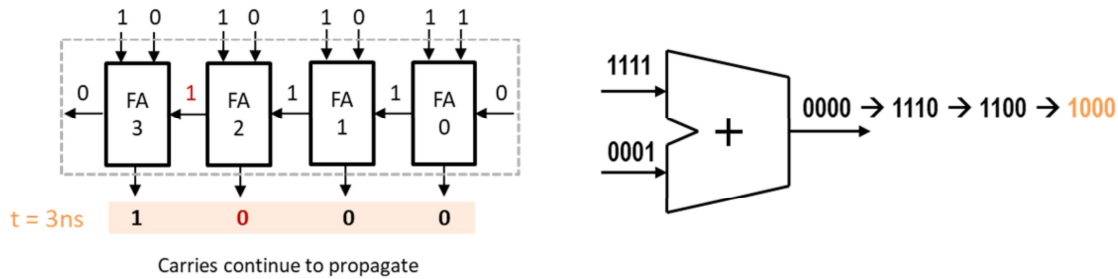
37

At $t = 2\text{ns}$, the carry output from **FA0** at $t = 1\text{ns}$ is now available as an input to **FA1**. This means that **FA1** will perform a new addition incorporating this carry. As a result, the output of **FA1** updates to **0** and the carry-out of **FA1** is now **1**, reflecting the impact of the delayed carry propagation.



Timing considerations

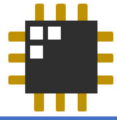
- We often want a circuit to be as fast as possible. However, it takes some time for signals to propagate (propagation delay)



* For simplicity, assume sum and carry-out are output by each FA after 1ns from input change

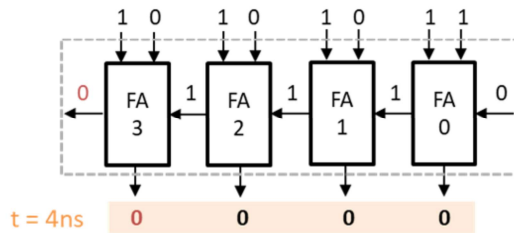
38

At $t = 3ns$, the carry output from **FA1** is now available and is added to the previous output of **FA2**. As a result, **FA2** updates its sum output to **0**, while its carry-out becomes **1**, continuing the propagation delay through the circuit.

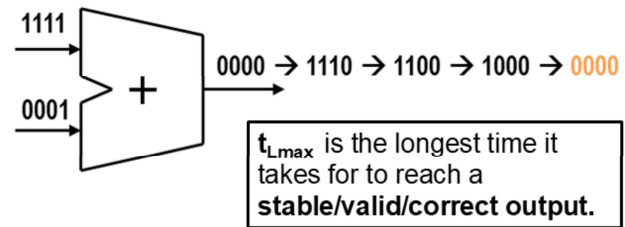


Timing considerations

- We often want a circuit to be as fast as possible. However, it takes some time for signals to propagate (propagation delay)



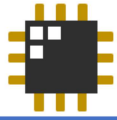
Now the sum is correct



* For simplicity, assume sum and carry-out are output by each FA after 1ns from input change

39

Finally, at $t = 4\text{ns}$, the carry output from **FA2** is added to the previous output of **FA3**, resulting in **FA3's** sum output becoming **0**. At this point, the circuit has fully stabilized, and we have reached the correct final output.



Timing considerations

- We always want to ensure that the output of a logic circuit is stable before we try to read the output value.
 - Stable output -> once all the logic delays have been resolved.
- This maximum logic delay, t_{Lmax} , would be relevant as we go into sequential circuits where the output is captured in periodic fashion.
 - We want to make sure the output is stable before we capture the output, which then defines the period at which we capture them.
 - This is exactly why processors, which are digital systems, have their maximum operating clock frequency as part of their specifications!

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Before reading a logic circuit's output, we need to make sure it has settled to the correct value. The output becomes stable once all delays in the circuit are finished. The longest delay, t_{Lmax} , is important in circuits that store values at set times. If the output isn't stable before saving it, errors can happen. This is why digital systems like processors have a maximum clock speed—to make sure everything updates correctly without mistakes.